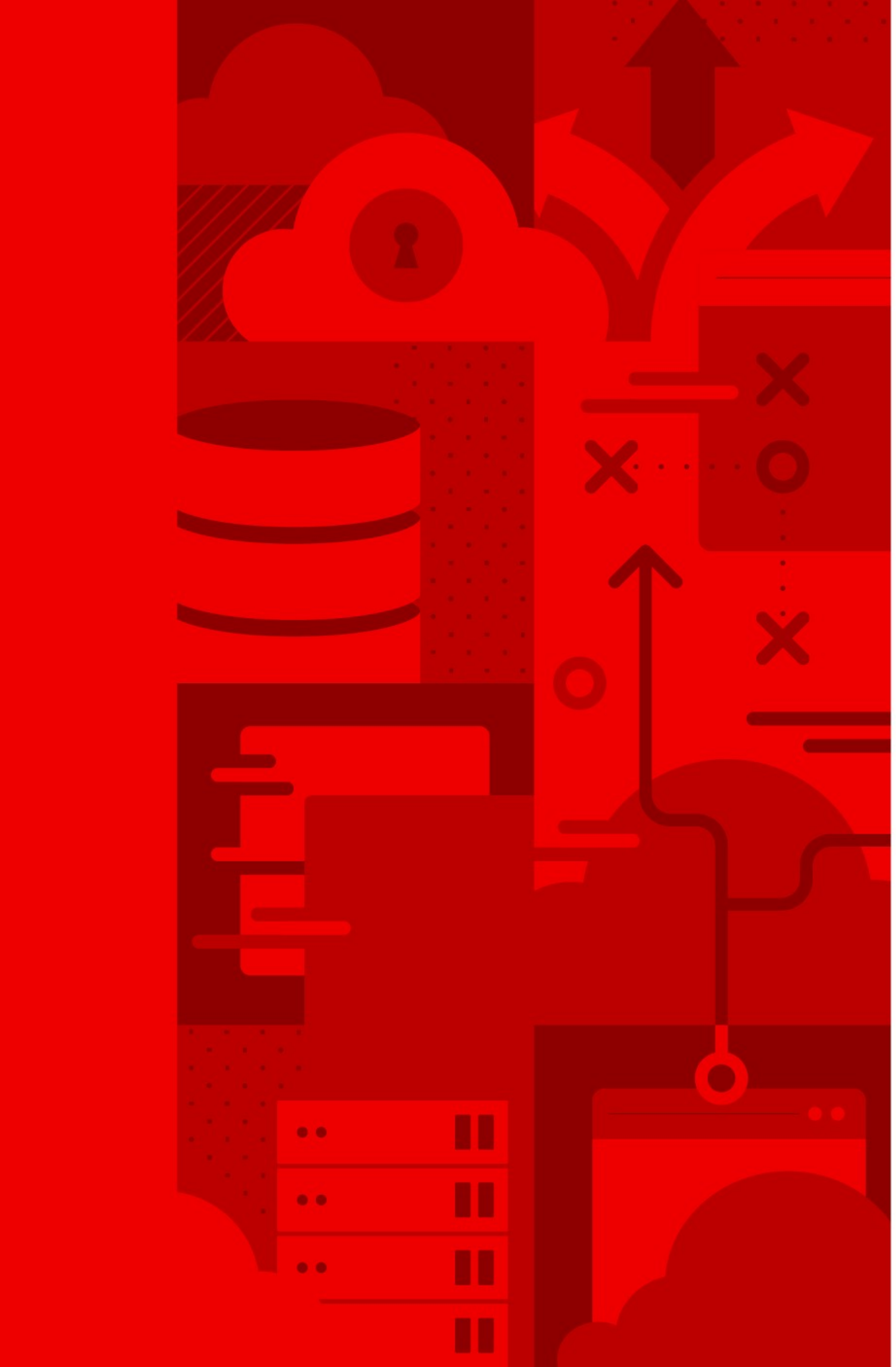

HANDS-ON WORKSHOP

Low-Latency Performance Workshop

OpenShift 4.20 — Achieving Microsecond-Level Performance for Mission-Critical Workloads

Tosin Akinosho

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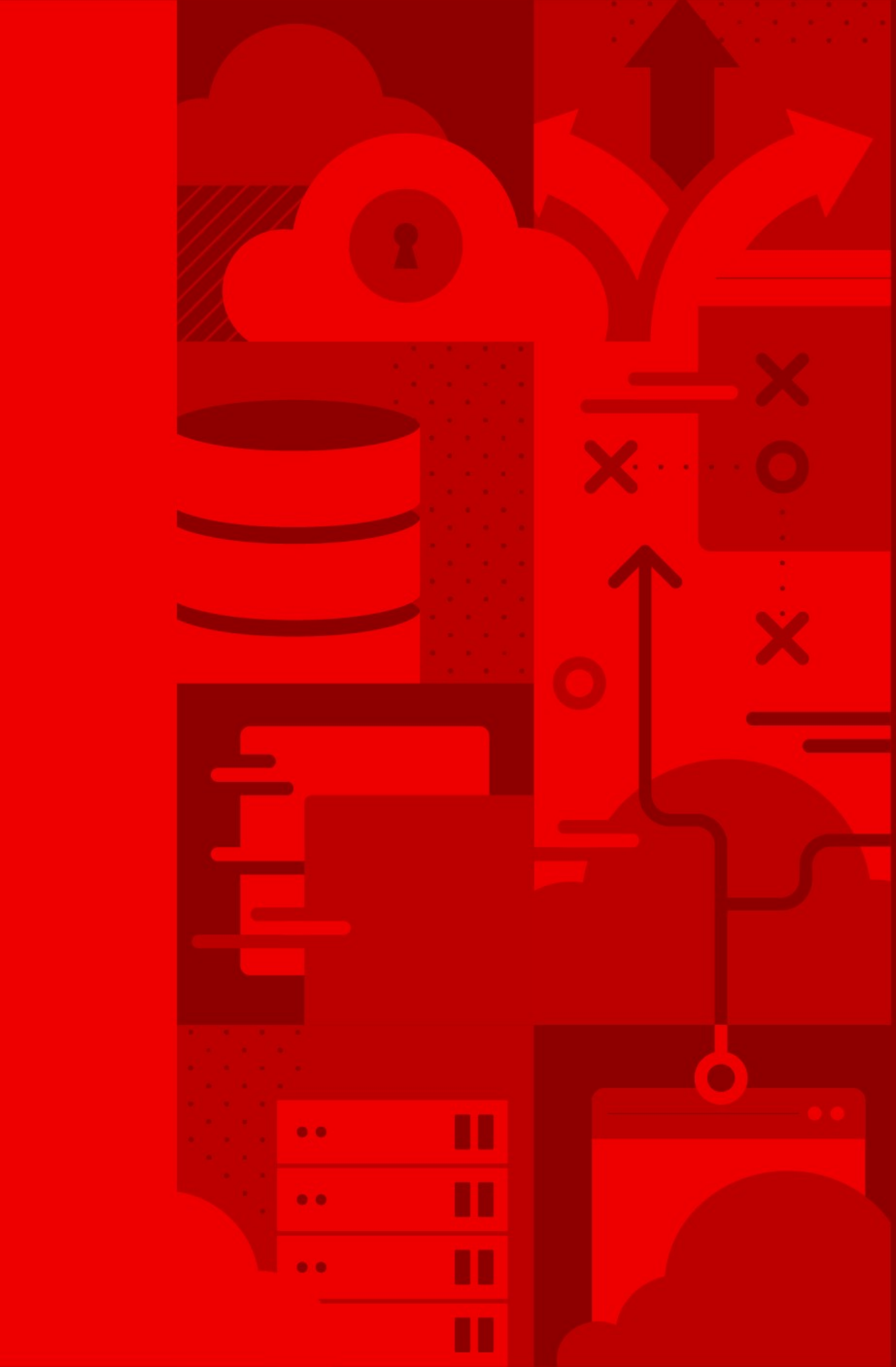

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The Latency Challenge

Why microseconds matter for your business

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THE CHALLENGE

Every Microsecond Has a Business Impact

- Financial trading platforms lose millions from millisecond delays — latency is a competitive differentiator.
- 5G and telecom networks require deterministic sub-millisecond response times for network function virtualization.
- Real-time analytics, IoT control systems, and gaming workloads demand predictable performance under load.
- Traditional container platforms introduce jitter from kernel scheduling, context switches, and shared resources.

⚡ **PERFORMANCE**
At 100,000 messages/second, properly tuned OpenShift achieves 2.3 µs P99 latency — nearly identical to bare-metal RHEL (2 µs).

Red Hat

TARGET INDUSTRIES

Who Needs Low-Latency Performance?

- Financial Services — HFT, risk engines, payment processing
- Telecommunications — 5G RAN, NFV, real-time call processing
- Gaming — real-time game servers, interactive streaming
- Industrial IoT — real-time control, edge analytics
- Real-Time Analytics — streaming data, fraud detection
- Scientific Computing — HPC, ML training, simulation

These industries share a common requirement: deterministic, predictable, microsecond-level response times.

01

Workshop Overview

What participants will learn and do

OVERVIEW

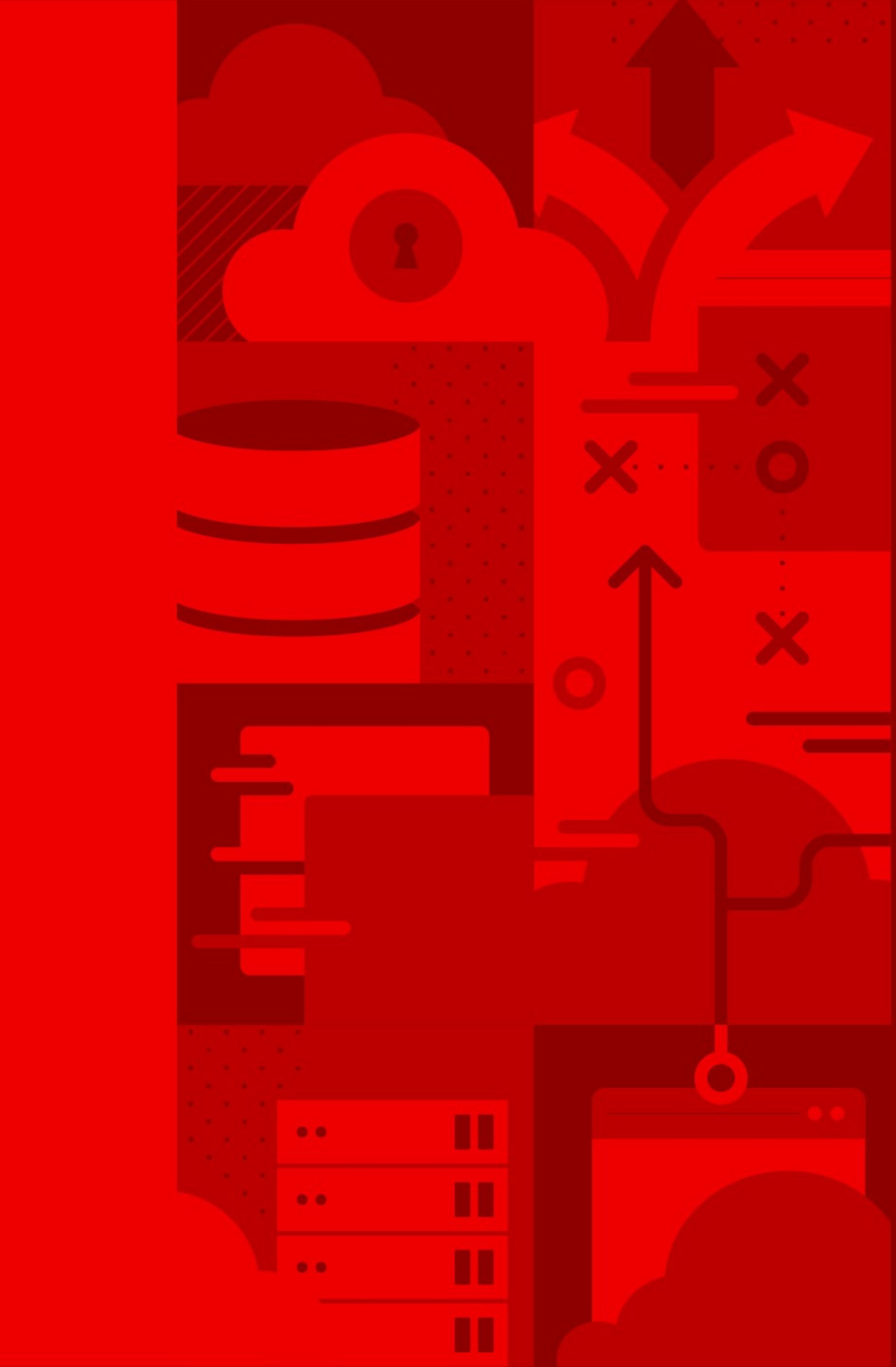
A Hands-On, Outcomes-Driven Workshop

- 8 progressive modules building from fundamentals to advanced optimization.
- Each participant gets a dedicated Single Node OpenShift (SNO) cluster on AWS.
- Real-world performance testing with kube-burner — quantitative before-and-after measurements.
- Covers containers AND virtual machines — a unified low-latency platform.
- Production-ready monitoring, alerting, and validation patterns included.

CURRICULUM

Workshop Modules

Mod 1	**Low-Latency Fundamentals**	Concepts, use cases, OpenShift performance architecture
Mod 2	**Environment Setup**	Verify SNO cluster, confirm operators, clone workshop repo
Mod 3	**Baseline Testing**	Establish performance baselines with kube-burner (P50/P95/P99)
Mod 4	**Core Performance Tuning**	CPU isolation, HugePages, RT kernel via Performance Profiles
Mod 5	**Low-Latency Virtualization**	OpenShift Virt with CPU pinning, HugePages, SR-IOV networking
Mod 6	**Monitoring & Validation**	Prometheus dashboards, alerting, continuous performance testing
Mod 7	**GPU Workloads (Optional)**	NVIDIA GPU Operator, GPU-enabled workloads, benchmarking
Mod 8	**Conclusion & Next Steps**	Key takeaways, production checklist, further learning



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Performance Results

Proven, measurable improvements

PROVEN RESULTS

Measurable Performance Improvements

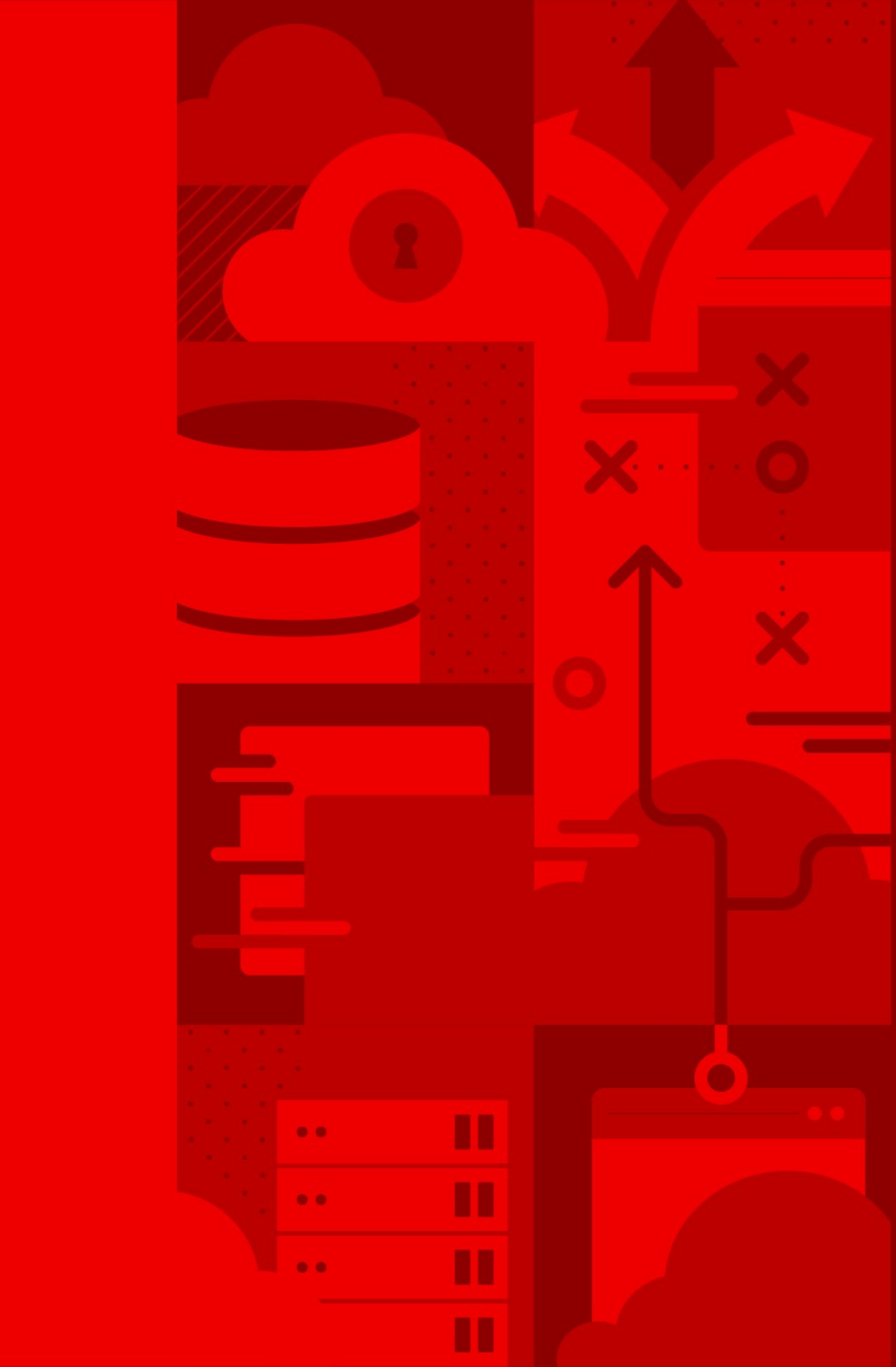
- OpenShift matches bare-metal RHEL P99 latency at 100K msg/s (2.3 µs vs 2 µs).
- At 1.4M messages/second, OpenShift sustains 2.8 µs P99 latency.
- A Tier 1 bank achieved 16× increase in daily transaction capacity over legacy platforms.
- Performance tuning yields 50–70% reduction in P99 tail latency.
- Workshop participants measure their own before-and-after improvements.

⚡ **PERFORMANCE**
Participants establish quantitative baselines in Module 3 and measure improvements after each optimization — every claim is verifiable.

TECHNOLOGY STACK

OpenShift 4.20 Performance Technologies

Node Tuning Operator	**System-level tuning**	Automated TuneD profiles, Performance Profile Controller, built-in to OCP 4.20
Performance Profiles	**Declarative optimization**	CPU isolation, HugePages, RT kernel, NUMA — single CR
SR-IOV Operator	**Hardware networking**	Direct NIC access bypassing kernel networking stack
OpenShift Virtualization	**VM optimization**	KubeVirt VMs with CPU pinning and dedicated resources
kube-burner 1.17+	**Performance testing**	Structured metrics with P50/P95/P99 percentile reporting
Prometheus + Grafana	**Monitoring stack**	Real-time dashboards, alerting, regression detection



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Architecture

How the workshop environment works

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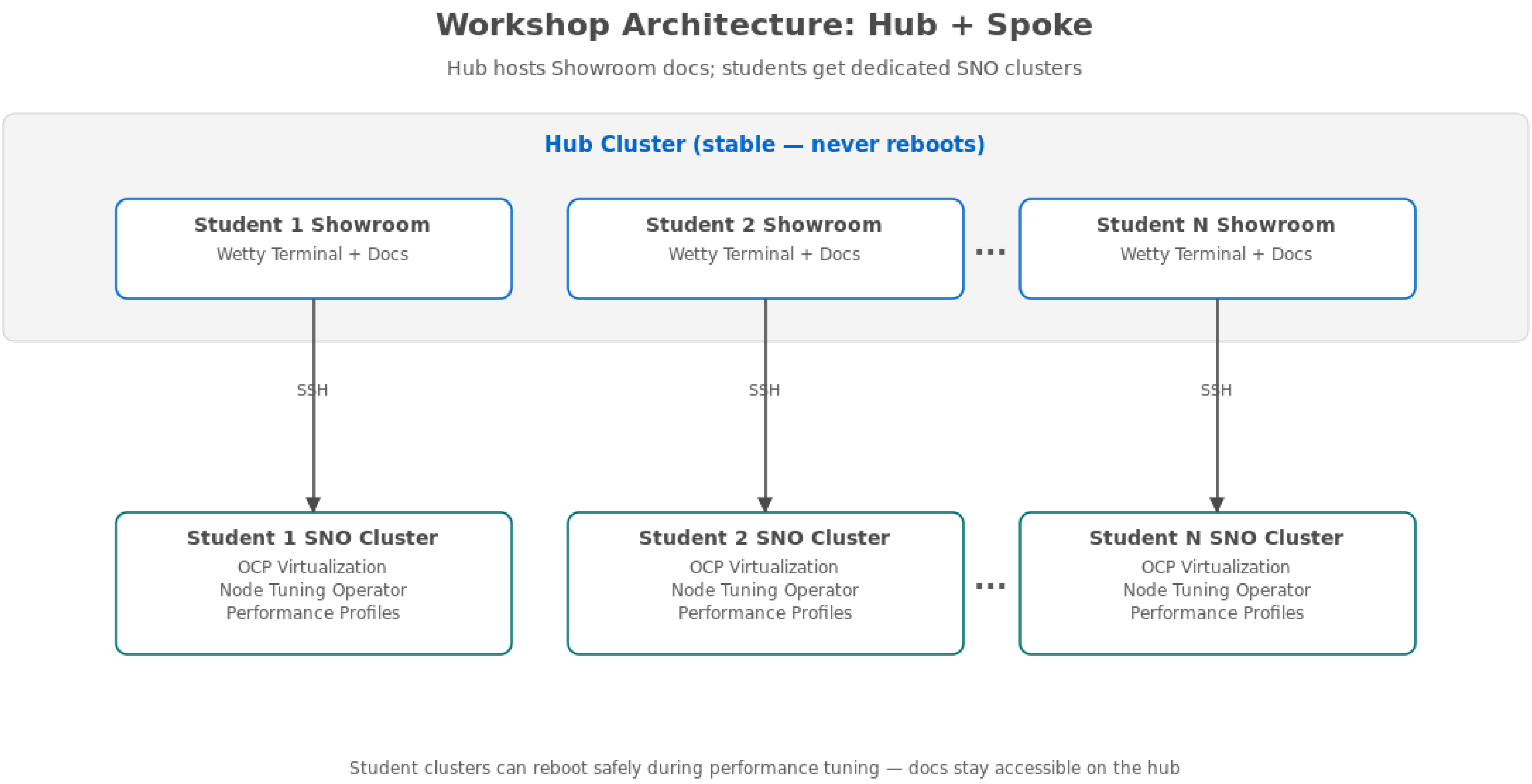
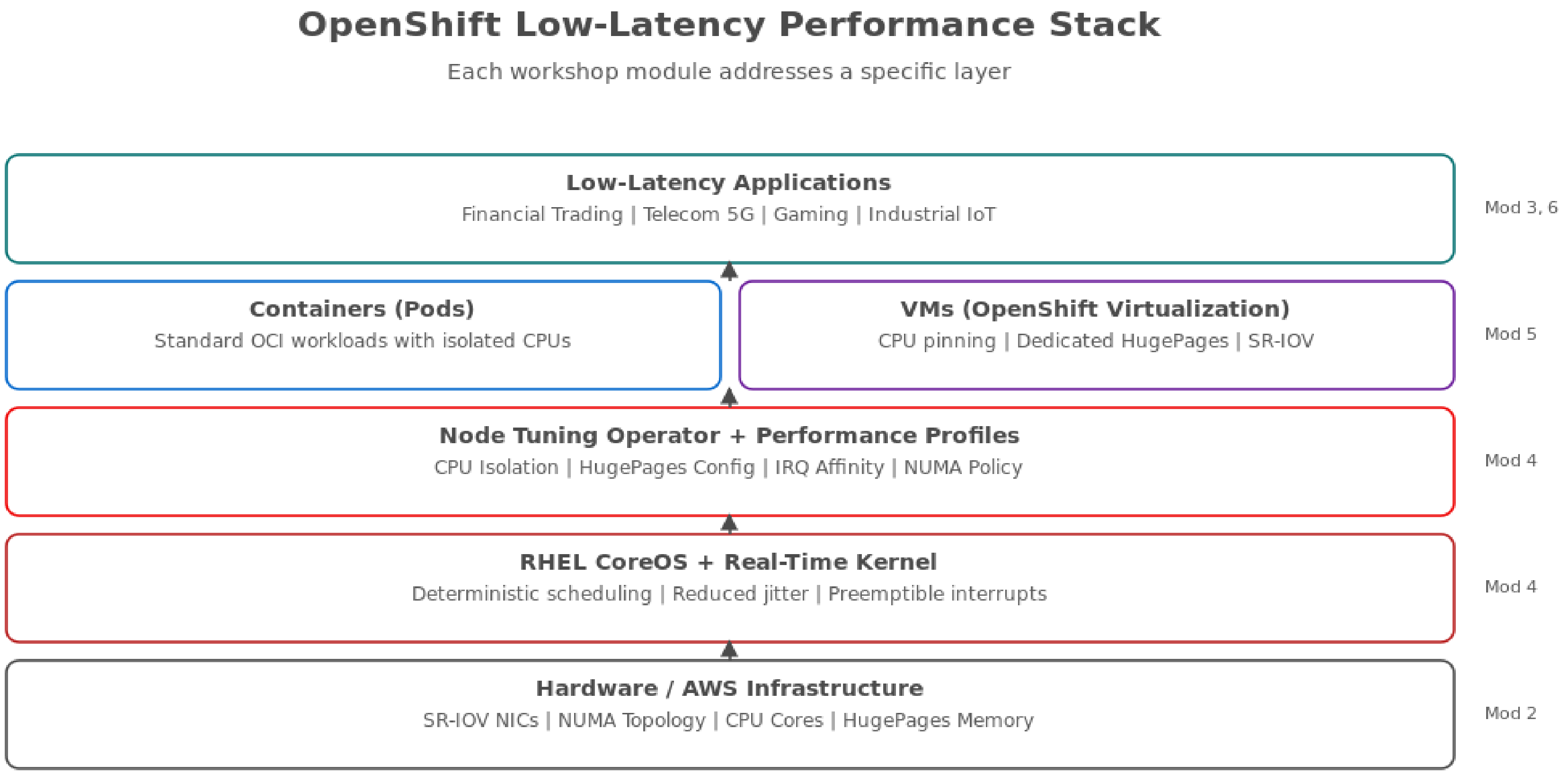
Hub + Spoke: Always-On Access During Cluster Reboots



Hub cluster hosts Showroom documentation; student SNO clusters can safely reboot during tuning exercises.

PERFORMANCE STACK

From Hardware to Application: The Optimization Layers

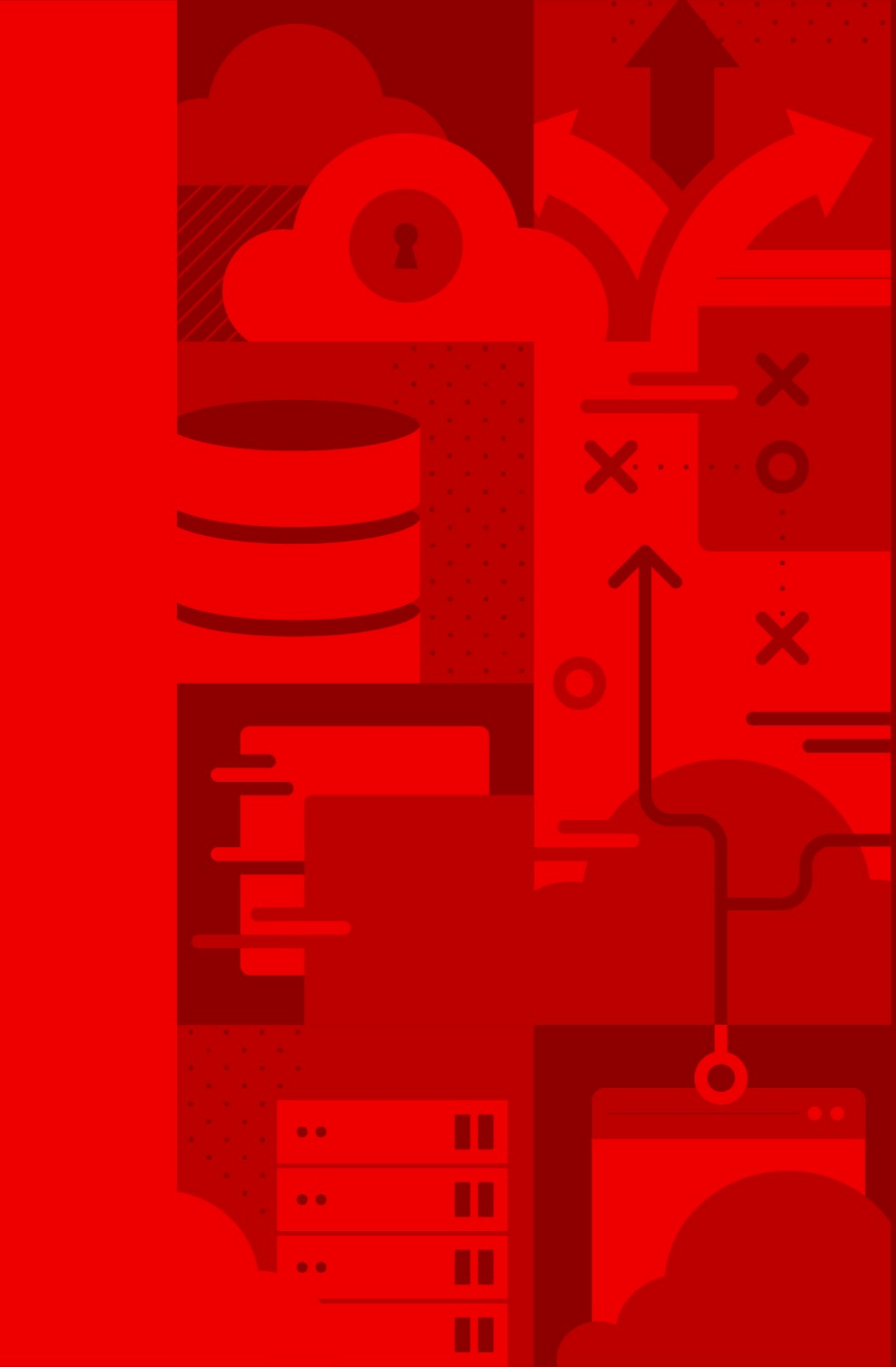


Each workshop module addresses a specific layer of the performance stack.

LAB ENVIRONMENT

Dedicated Resources for Every Participant

- Dedicated Single Node OpenShift (SNO) cluster on AWS (m5.4xlarge — 16 vCPUs, 64 GB RAM).
- Personal Showroom instance with embedded terminal on the hub cluster.
- Pre-installed operators: OpenShift Virtualization, SR-IOV Network Operator, Node Tuning Operator.
- Bastion host for cluster management with pre-configured oc CLI and KUBECONFIG.
- Full cluster-admin privileges — unrestricted hands-on exploration.
- All infrastructure provisioned and torn down automatically via AgnosticD.



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Deep Dive Highlights

Key hands-on exercises and techniques

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MODULE 4 HIGHLIGHT

Core Performance Tuning: CPU Isolation and HugePages

- Create Performance Profiles that isolate specific CPU cores for workloads — eliminating interference from system processes.
- Configure 2 MB and 1 GB HugePages to reduce TLB misses and memory management overhead.
- Apply real-time kernel tuning for deterministic scheduling and reduced jitter.
- Measure before-and-after performance with kube-burner — quantitative proof of improvement.

⚡ **PERFORMANCE**
A single Performance Profile CR manages CPU isolation, HugePages, RT kernel, and NUMA awareness — declarative, version-controlled, and reproducible.

Performance Profile Custom Resource

yaml · OpenShift 4.20

```yaml
# Performance Profile for low-latency workloads
apiVersion: performance.openshift.io/v2
kind: PerformanceProfile
metadata:
  name: low-latency-profile
spec:
  cpu:
    isolated: "4-15"       # Cores for workloads
    reserved: "0-3"        # Cores for OS/platform
  hugepages:
    defaultHugepagesSize: 1G
    pages:
      - size: 1G
        count: 4
  realTimeKernel:
    enabled: true
  numa:
    topologyPolicy: single-numa-node
```

Participants create, apply, and validate this profile — then measure the performance difference.

MODULE 5 HIGHLIGHT

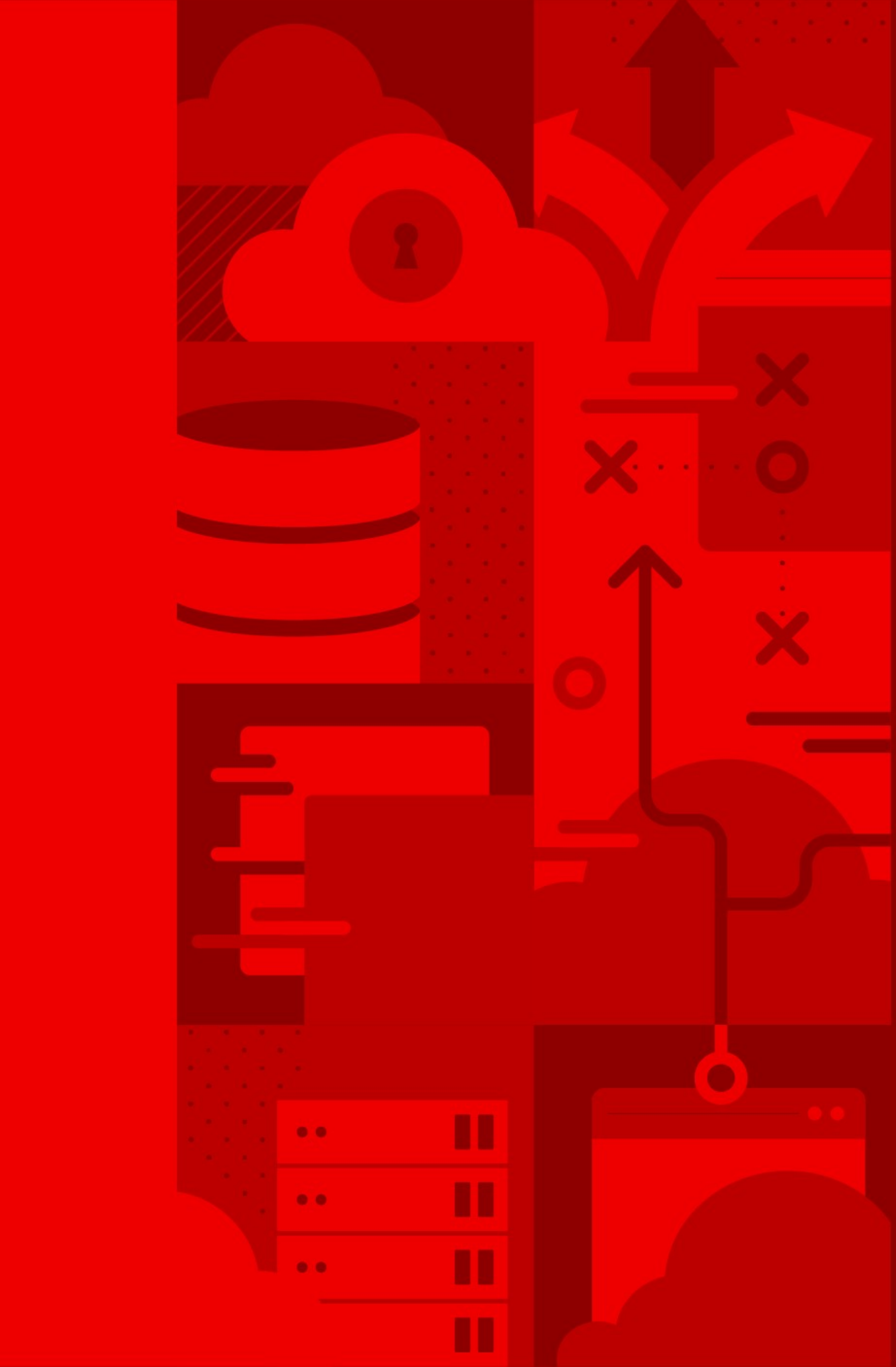
Low-Latency Virtual Machines on OpenShift

- Run VMs alongside containers on the same cluster — unified management and security.
- Assign dedicated CPUs and HugePages to VMs for deterministic performance.
- Configure SR-IOV networking for direct NIC access — bypassing the kernel networking stack.
- Measure VMI startup latency and network performance with kube-burner.
- Compare container vs. VM performance side by side with analysis scripts.

MODULE 6 HIGHLIGHT

Production-Ready Monitoring and Validation

- Deploy Prometheus and Grafana dashboards purpose-built for latency monitoring.
- Create alerts for performance regressions and SLA threshold violations.
- Validate optimizations across containers, VMs, and networking layers.
- Implement continuous performance testing workflows with CronJobs for ongoing assurance.
- Detect performance regressions before they reach production users.



05

Getting Started

Prerequisites, audience, and logistics

AUDIENCE

Who Should Attend?

- Platform Engineers responsible for OpenShift cluster performance and SLA compliance.
- SREs managing production clusters with latency-sensitive workloads.
- Performance Specialists conducting profiling, testing, and optimization.
- DevOps Engineers implementing GitOps for performance-critical infrastructure.
- Solutions Architects designing OpenShift for low-latency use cases.
- *Recommended: basic OpenShift/Kubernetes experience*

LOGISTICS

Prerequisites and Workshop Format

- Basic understanding of OpenShift and Kubernetes concepts (no advanced expertise required).
- Familiarity with Linux command line and YAML configuration.
- All infrastructure is pre-provisioned — no setup required from participants.
- Workshop duration: approximately one full day (8 modules, progressive complexity).
- All exercises use the oc CLI and web console — no special tools needed on participant machines.

Optional GPU module (Module 7) requires GPU-enabled nodes and can be skipped.

NEXT STEPS

Ready to Optimize Your Latency?

- Contact your Red Hat account team to schedule the workshop.
- Workshop can be delivered on-site, remotely, or as a self-paced lab.
- Participants leave with hands-on experience, documented results, and a production optimization checklist.
- All workshop materials, scripts, and configurations are open source and reusable.

⚡ PERFORMANCE
From baseline to optimized in one day — every participant leaves with quantitative proof that OpenShift delivers bare-metal-class latency.